PORTFOLIO MANAGEMENT

Complete Combined Study Notes

Layman's Language | Full Formulas | Real-Life Examples | Exam-Ready

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Unit I: Introduction to Portfolio Management
Unit II: Portfolio Optimization
Unit III: Capital Market Theory & Pricing Models

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UNIT I — Introduction to Portfolio Management

1. Asset

An asset is any valuable economic entity from which future economic benefits are expected to flow to the owner. The owner legally acquired it as a result of past transactions.

Types of Assets

- Physical / Tangible Assets: Things you can touch — real estate, machinery, furniture, gold. Also called capital assets in accounting.
- Intangible Assets: Things you cannot touch — goodwill, patents, brand name, competitive knowledge, franchise rights.
- Financial / Liquid Assets: Paper or digital assets — shares, bonds, mutual funds, currency. Gold & precious metals are both tangible AND liquid.

Real-Life Example — Asset Types

Tangible: A steel plant owns blast furnaces worth ₹500 Cr. These are capital assets generating future revenue.

Intangible: Infosys's brand value (goodwill) or a pharmaceutical patent on a cancer drug.

Financial: Your SBI FD (bond-like), 100 shares of Reliance (equity), or a gold ETF on NSE.

Personal: A flat you own (physical asset) + mutual fund holdings (financial asset) + brand recognition of your startup (intangible asset) together form your personal 'asset base'.

2. Return — How Much Did Your Investment Grow?

Return measures how much an asset has gained (or lost) relative to its starting value. It is always expressed as a rate (percentage). Since the future price is uncertain, return is treated as a random variable.

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Return = [Closing Price (current) − Closing Price (previous) + Dividend] / Closing Price (previous)
r_i = [V_i(T) − V_i(0)] / V_i(0)
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We work with the expected value (mean): $\mu_i = E(r_i)$ and variance $\sigma_i^2 = Var(r_i)$.

Real-Life Example — Return Calculation

You buy 1 share of TCS at ₹3,500 on 1 Jan. By 31 Dec it is ₹4,025 and paid ₹50 dividend.

Return = (4025 − 3500 + 50) / 3500 = 575/3500 ≈ 16.4%

This 16.4% is the realised return. Before the year ends, it is a random variable — you don't know if TCS will rise or fall.

3. Risk — The Uncertainty of Returns

Risk is the degree of uncertainty of return on an asset — the probability of a financial loss or deviation from expected return.

Risk Level	Description	Example
Risk = 0	Risk-free asset. Future value known with certainty.	Government T-bills, G-Sec
Risk > 0	Risky asset. Future value is uncertain.	Equities, mutual funds, crypto

Two Main Types of Risk

Feature	Unsystematic Risk	Systematic Risk
Definition	Unique to a company or sector	Common to ALL businesses
Examples	Factory fire, CEO scandal, product recall at ONE company	RBI rate hikes, inflation, recession, pandemics
Can be diversified?	YES ✅— Eliminated by diversification	NO ❌— Cannot be diversified away
Measurement	Residual variance ($\sigma^2\varepsilon$)	Beta (β)

Real-Life Example — Types of Risk

Unsystematic: In 2019, IL&FS defaulted. Investors holding ONLY IL&FS bonds were wiped out. Those holding a diversified mix (bonds + equities + gold) were partially protected.

Systematic: During COVID-19 (March 2020), EVERY stock fell — Reliance, TCS, HDFC Bank, Infosys — regardless of company quality. No diversification could save you from that systemic shock.

4. Short Selling

Short selling means selling an asset you do NOT currently own, anticipating its price will fall. You borrow the asset, sell it at today's price, and plan to buy it back cheaper later.

- If units of asset i held > 0 → Long Position (normal buying)
- If units of asset i held < 0 → Short Position (short selling)
- Asset weight $w_i < 0$ mathematically represents a short position

Real-Life Example — Short Selling

You believe Zomato's share at ₹200 is overvalued. You borrow 100 shares and sell → receive ₹20,000.

3 months later Zomato falls to ₹140. You buy 100 shares for ₹14,000 and return them.
Profit = ₹20,000 − ₹14,000 = ₹6,000 (minus borrowing fees).

□ Risk: If Zomato ROSE to ₹260, you would LOSE ₹6,000 — short selling can have unlimited losses!

5. Model Assumptions

The mathematical portfolio model is built on these simplifying assumptions:

- Prices of all assets are strictly positive at any point in time (past, current, future).
- Return on any asset is a random variable taking finite values.
- Divisibility: An investor can own a fraction of an asset (possible via mutual fund units).
- Liquidity: Any quantity of an asset can be bought or sold on demand at the market price.
- No transaction costs or commissions (simplifying assumption).

□ Key Concept

Divisibility Example: You cannot buy 0.3 shares of MRF (₹1.4 lakh/share). But a mutual fund lets you invest ₹5,000 and own a fraction of a portfolio containing MRF.

Liquidity Example: A 91-day T-bill can be sold on demand; a 10-acre farm plot cannot — it is illiquid.

6. Portfolio — Don't Put All Eggs in One Basket

A portfolio is a collection of two or more assets $\{a_1, a_2, ..., a_n\}$ represented by an ordered n-tuple:

$$\theta = (x_1, x_2, ..., x_n) \quad \text{where } x_i \in \mathbb{R}, \quad i = 1, 2, ..., n$$

x_i = number of units of asset a_i owned by the investor (negative means short position).

6.1 Asset Weight (w_i)

The weight w_i of asset a_i is the percentage of total investment allocated to that asset at t = 0.

$$w_i = V_i(0) \cdot x_i \;/\; \Sigma V_i(0) \cdot x_i = \text{Amount invested in asset } i \;/\; \text{Total amount invested}$$

$$\text{Constraint: } \Sigma w_i = w_1 + w_2 + ... + w_n = 1$$

Real-Life Example — Portfolio Weights

Portfolio: ₹1,00,000 total. ₹40,000 in HDFC Bank, ₹35,000 in Infosys, ₹25,000 in Gold ETF.

w_HDFC = 0.40, w_Infosys = 0.35, w_Gold = 0.25
Check: 0.40 + 0.35 + 0.25 = 1.00 ✓

If you short Infosys worth ₹10,000: w_Infosys = −0.10 (negative weight = short position).

6.2 Portfolio Return & Variance

Expected Return of Portfolio — the weighted average of individual expected returns:

$$\mu_P = \Sigma\, w_i \cdot \mu_i = w_1\mu_1 + w_2\mu_2 + \ldots + w_n\mu_n \qquad \text{(Matrix form: } \mu_P = m^T w\text{)}$$

Variance (Risk) of Portfolio — accounts for co-movement between all pairs of assets:

$$\sigma^2_P = \Sigma_i\, w_i^2 \cdot \sigma_i^2 + 2 \cdot \Sigma_{i<j}\, w_i \cdot w_j \cdot \sigma_{ij} \qquad \text{(Matrix form: } \sigma^2_P = w^T C w\text{)}$$

where C is the n×n variance-covariance matrix: $C_{ii} = \sigma_i^2$ (variance), $C_{ij} = \sigma_{ij}$ (covariance).

Why Two Parts in Portfolio Variance?

Part 1: Sum of individual variances (each asset's own risk, weighted by w^2)
Part 2: Cross-terms — how PAIRS of assets move together (covariance terms)

If assets move in opposite directions, the cross-terms REDUCE total risk.
This is the mathematical magic of diversification!

Real-Life Example — Portfolio Variance

Two stocks: HDFC Bank (μ_1 =12%, σ_1 =15%) and Infosys (μ_2 =16%, σ_2 =20%). Weights: 60/40. ρ=0.3.

Expected return = 0.6×12 + 0.4×16 = 13.6%
$\sigma_{12} = \rho \cdot \sigma_1 \cdot \sigma_2$ = 0.3 × 15 × 20 = 90
Portfolio Variance = $(0.6)^2 \times (15)^2 + (0.4)^2 \times (20)^2$ + 2×0.6×0.4×90
= 0.36×225 + 0.16×400 + 43.2 = 81 + 64 + 43.2 = 188.2
Portfolio Std Dev = √188.2 ≈ 13.7% ← Less than Infosys's own 20%! Diversification works.

7. Diversification — The Only Free Lunch in Finance

Diversification means spreading investments across different assets so poor performance of one is offset by good performance of others. It reduces unsystematic risk WITHOUT reducing expected return.

Mathematical Impact of Correlation (ρ) on Portfolio Risk

$$\text{Two-Asset Portfolio Variance:} \quad \sigma^2\_P = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$$

Correlation ρ	Effect on Risk	Can Zero-Risk Portfolio Exist?
$\rho = +1$	No benefit. Risk is a weighted average. Assets move in perfect sync.	No
$-1 < \rho < +1$	Partial risk reduction. More negative = better diversification.	No
$\rho = -1$	Maximum diversification. Assets move in exactly opposite directions.	YES — theoretically possible

Minimum Variance Portfolio when ρ = −1

$$\text{Optimal weight for asset 1:} \quad w^* = \sigma_2 / (\sigma_1 + \sigma_2)$$

At this weight, the portfolio variance = 0 — a RISK-FREE portfolio from two risky assets!

Minimum Variance Portfolio: General Case (−1 < ρ < 1)

$$s\_min = (\sigma_1^2 - \rho \cdot \sigma_1 \cdot \sigma_2) / (\sigma_1^2 + \sigma_2^2 - 2\rho \cdot \sigma_1 \cdot \sigma_2)$$

$$\sigma^2\_min = \sigma_1^2 \sigma_2^2 (1-\rho^2) / (\sigma_1^2 + \sigma_2^2 - 2\rho \cdot \sigma_1 \cdot \sigma_2)$$

□ Real-Life Example — Diversification

Gold and Nifty 50 tend to move in OPPOSITE directions (when markets crash, gold rises).
Combining them reduces overall portfolio risk.
Many advisors recommend 10–15% gold allocation for this very reason.

Imagine 5 businesses: restaurant, umbrella shop, sunscreen company, hospital, online tutor.
• Sunny days: sunscreen & restaurant thrive; umbrella suffers.
• Rainy days: umbrella thrives; sunscreen suffers.
Investing in all 5 smooths out your overall returns — that's diversification!

8. Alternate Risk Measures

Beyond standard variance, several other measures capture risk more precisely for different situations.

8a. Mean Absolute Deviation (MAD)

Instead of squaring deviations (variance), MAD uses the absolute value. It is simpler to compute and less sensitive to extreme outliers.

$$MAD = E[\ |R - E(R)|\ ]$$

8b. Semi-Variance (Downside Risk)

Variance penalises BOTH upside and downside deviations from expected return. But investors don't mind getting MORE than expected! Semi-variance only penalises DOWNSIDE deviations.

$$\text{Semi-Variance} = E[(R - E)^{-}]^{2} \quad \text{where} \quad (R-E)^{-} = (R-E) \text{ if } R<E, \text{ else } 0$$

Real-Life Example — Semi-Variance

Expected return = 10%. Month 1 you get 15% (good!). Month 2 you get 5% (bad!).

• Variance penalises BOTH months equally.

• Semi-variance ONLY penalises Month 2 (the shortfall).

This better matches how real investors think about risk.

8c. Value at Risk (VaR)

VaR is the MAXIMUM loss you can expect, at a given confidence level, over a specific time period, under normal market conditions.

$$\text{VaR}(\alpha) = \text{Number above which there is only } (1-\alpha)\% \text{ probability of exceeding as a loss}$$

Real-Life Example — VaR

Portfolio = ₹10,00,000. Manager says: '1-year VaR at 99% confidence = ₹1,00,000'.

Meaning: Under normal conditions, there is 99% chance your loss will NOT exceed ₹1 lakh in 1 year.

There is a 1% chance your loss COULD exceed ₹1 lakh.

Every bank and mutual fund in India uses VaR to report risk to SEBI. You'll see it in all fund factsheets.

8d. Conditional Value at Risk (CVaR) — Expected Shortfall

CVaR asks: 'If things DO go wrong (beyond VaR), how bad will it be on AVERAGE?' It is the expected loss GIVEN that the loss exceeds VaR. CVaR always gives a more complete picture of tail risk.

Real-Life Example — CVaR

VaR says: 'You'll lose at most ₹1 lakh, 99% of the time.'

CVaR says: 'On that 1% of bad days, your AVERAGE loss will be ₹1.8 lakh.'

CVaR answers the 'how bad is worst case?' question that VaR ignores.

Risk Measure	What it Measures	Use Case
Variance (σ^2)	Average squared deviation from mean	Standard Markowitz optimization

MAD	Average absolute deviation from mean	Linear programming approach
Semi-Variance	Only downside deviations penalised	Investor cares only about losses
VaR	Maximum loss at given confidence level	Regulatory reporting (SEBI, RBI)
CVaR	Expected loss beyond VaR (tail risk)	Best for capturing extreme losses

9. Liquidity and Market Impact

Liquidity refers to how easily and quickly an asset can be bought or sold at its fair market price WITHOUT significantly affecting that price.

- High Liquidity: Nifty 50 stocks, Government bonds — buy/sell large quantities instantly.
- Low Liquidity: Small-cap stocks, real estate, art — selling quickly forces accepting a lower price.

Market Impact: When you trade a LARGE quantity, your own trade moves the price against you. This cost is called market impact. It's why big mutual funds can't just 'dump' stocks all at once.

Real-Life Example — Market Impact

HDFC Mutual Fund needs to sell ₹500 crore of a small-cap stock.
If they sell it all at once, the stock price crashes.
So they sell in small lots over days/weeks to minimize market impact.

10. Hedging Principle

Hedging means taking an offsetting position to reduce or eliminate risk from an existing investment. It's like buying insurance for your portfolio.

Real-Life Example — Hedging

You hold 100 shares of TCS worth ₹10 lakh. You fear the market will fall.
You BUY a put option (right to sell TCS at today's price).
If TCS falls by 20%, your put option gains value — offsetting your loss on the shares.

You paid a small premium (like an insurance premium) to protect your downside.
Hedging doesn't maximise profit — it LIMITS LOSS.

UNIT II — Portfolio Optimization

11. Multi-Asset Portfolio Optimization

For n assets, the full optimization framework uses matrix algebra. Key notation:

- w = column vector of weights (w_1 , w_2 , ..., w_n)
- m = vector of expected returns (μ_1 , μ_2 , ..., μ_n)
- C = n×n variance-covariance matrix
- e = vector of ones (for full investment constraint)

11.1 Global Minimum Variance Portfolio (MVP)

The portfolio with the LEAST possible risk among all fully invested portfolios:

$$\text{Minimize } \sigma^2\_P = w^T C w \quad \text{Subject to: } e^T w = 1$$

$$\text{Solution: } w^* = C^{-1}e \,/\, (e^T C^{-1} e)$$

Key Concept

C^{-1} = Inverse of the covariance matrix.
$e^T C^{-1} e$ = a scalar (sum of all elements of C^{-1}).

Special case (uncorrelated assets): $w_i = (1/\sigma_i^2) / \Sigma(1/\sigma_j^2)$
→ Put more weight in lower-variance assets.

Real-Life Example — Global MVP

3 uncorrelated assets: σ_1^2=5%, σ_2^2=10%, σ_3^2=20%.
$1/\sigma_1^2 = 20$, $1/\sigma_2^2 = 10$, $1/\sigma_3^2 = 5$. Sum = 35.
w_1 = 20/35 ≈ 57%, w_2 = 10/35 ≈ 29%, w_3 = 5/35 ≈ 14%

Put the MOST money in the least risky asset.
Like a cautious investor: 57% in FD, 29% in balanced fund, 14% in small-cap equity.

12. Markowitz Portfolio Selection Model & Efficient Frontier

Harry Markowitz (1952, Nobel Laureate) proposed the first mathematical framework for optimal portfolios. The central insight: don't just pick assets with high return — consider the COMBINATION of returns AND risks and how they interact.

The Optimization Problem

For every target return level μ_0 , find the portfolio with MINIMUM variance:

$$\text{Minimize: } w^T C w$$

$$\text{Subject to: } m^T w = \mu_0 \quad \text{(return constraint)} \quad \text{AND} \quad e^T w = 1 \quad \text{(full investment constraint)}$$

Solution Using Lagrange Multipliers

$$w = (\alpha m^T + \beta e^T) \cdot C^{-1} / 2$$

$$\text{Full solution: } w^* = [\det(\mu_0, m^T C^{-1} e) \cdot C^{-1} m + \det(m^T C^{-1} m, \mu_0) \cdot C^{-1} e] / \det(m^T C^{-1} m, m^T C^{-1} e)$$

The Efficient Frontier is the UPPER half of the Markowitz curve — portfolios where higher risk gives higher return. The lower half contains dominated portfolios (same risk, less return).

Real-Life Example — Efficient Frontier

Think of it like a menu at a restaurant:
• Every point on the frontier = 'best value meal' — for a given price (risk), max flavour (return).
• Points BELOW the frontier = 'bad deals' — same price but less flavour.

Example: At σ=15%, you can earn μ=12% (efficient) or only μ=9% (dominated/inefficient).
A rational investor always chooses the efficient option.

13. Two-Fund Theorem

A fundamental result: ANY efficient portfolio can be created by combining just TWO specific efficient portfolios (funds). You don't need to solve the full optimization for every investor.

$$w\_P = \alpha \cdot w_1 + (1-\alpha) \cdot w_2 \quad \text{for some scalar } \alpha$$

Why This Matters

Practical Implication: An investment manager needs to offer only TWO well-chosen portfolios.
Every investor, regardless of risk appetite, can combine these two to get their desired portfolio.

This is why Index Funds + a Money Market Fund can span the entire efficient frontier for retail investors!
This also underpins the logic of 'balanced' mutual fund options.

Real-Life Example — Two-Fund Theorem

Fund X = Nifty 50 Index Fund (μ=13%, σ=18%) — approximates the market portfolio.
Fund Y = Overnight Liquid Fund (μ=6.5%, σ≈0%) — risk-free proxy.

Conservative investor: 20% in X + 80% in Y → μ ≈ 7.8%, low risk.
Aggressive investor: 80% in X + 20% in Y → μ ≈ 11.7%, higher risk.

Both investors are on the efficient frontier, combining the SAME two funds!

14. Alternate Risk Measure Models for Optimization

When variance is replaced by alternate risk measures, the optimization problem changes structure.

14a. Mean Absolute Deviation (MAD) Model

Converts a quadratic optimization problem (hard) into a linear program (easy to solve).

$$\text{Minimize: } (1/T)\ \Sigma_t\ |\Sigma_i\ w_i r_{it} - \mu|$$

$$\text{Subject to: } \Sigma w_i = 1,\quad \Sigma w_i \mu_i \geq \mu_0,\quad w_i \geq 0$$

14b. Mean Semi-Absolute Deviation Model

Like MAD but only counts NEGATIVE deviations (below expected return). Focuses purely on downside risk.

$$\text{Minimize: } E[(R - E(R))^-] \quad \text{where} \quad (x)^- = \max(0, -x)$$

14c. Mean Value at Risk (VaR) Model

Minimize the VaR of the portfolio subject to a target return constraint. Captures tail risk but is more complex to optimize.

14d. Mean Conditional VaR (CVaR) Model

Minimize the Expected Shortfall (CVaR). Considered the BEST risk measure because it accounts for extreme losses. Also convex — easier to optimize than VaR.

Real-Life Comparison — Risk Measures as Travel Time Analogy

Measuring risk of your daily commute to work:
- Variance: Average squared deviation from expected travel time.
- MAD: Average absolute deviation from expected travel time.
- Semi-Variance/MAD: Only count days you were LATE (downside only).
- VaR: Worst-case travel time on 99% of days.
- CVaR: Average travel time on those 1% worst (extreme) days.

15. Marginal Contributions to Risk (MCR) & Implied Returns

Each asset contributes a specific amount to the portfolio's total risk. MCR measures how much the portfolio's risk changes when you add a tiny bit more of a particular asset.

$$MCR_i = \partial\sigma\_P / \partial w_i = (Cw)_i / \sigma\_P$$

Implied Returns: Given a benchmark portfolio, we can work backwards to find 'implied expected returns' — what returns the market is pricing in, given current weights and covariances. This is used in the Black-Litterman model.

Real-Life Example — MCR

A risk manager at HDFC Bank looks at the Nifty 50.
Each stock contributes differently to the index's overall risk.
Stocks with high MCR are 'risky contributors' and should be reduced.
Stocks with low MCR can be increased to diversify better.

16. Portfolio Performance Evaluation Measures

Once a portfolio is built, how do you evaluate if your fund manager is doing a good job? Three classic measures help — all based on CAPM.

16a. Sharpe Ratio

Measures return earned PER UNIT of total risk (standard deviation). Higher = better. Developed by William Sharpe (Nobel Laureate).

$$\text{Sharpe Ratio} = (\text{Portfolio Return} - \text{Risk-Free Rate}) / \text{Portfolio Standard Deviation}$$

$$S\_P = (\mu\_P - \mu\_f) / \sigma\_P$$

Real-Life Example — Sharpe Ratio

Portfolio A: Return=18%, σ=20%, μ_f=7%. S_A = (18−7)/20 = 11/20 = 0.55
Portfolio B: Return=12%, σ=8%, μ_f=7%. S_B = (12−7)/8 = 5/8 = 0.625

Portfolio B has a HIGHER Sharpe Ratio even though it earned LESS absolute return!
It earned more return per unit of risk. B is the better risk-adjusted performer.

Think of Sharpe Ratio as: 'How many rupees of excess return per rupee of risk?'

16b. Treynor Ratio

Like Sharpe but uses SYSTEMATIC RISK (beta, β) instead of total risk. Better for comparing funds that are part of a larger diversified portfolio.

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Treynor Ratio  =  (Portfolio Return − Risk-Free Rate)  /  Beta (β)
T_P  =  (μ_P − μ_f)  /  β_P
```

Real-Life Example — Treynor Ratio

Portfolio A: Return=18%, β=1.8, μ_f=7%. $T_A = (18-7)/1.8 = 6.11$
Portfolio B: Return=12%, β=0.7, μ_f=7%. $T_B = (12-7)/0.7 = 7.14$

Portfolio B wins! More excess return per unit of market (systematic) risk.
Use Treynor when you're ADDING a fund to an already diversified portfolio.
Total σ doesn't matter as much — only systematic risk counts.

16c. Jensen's Alpha (α)

Measures how much a fund OUTPERFORMED (or underperformed) what CAPM predicts. Positive alpha = fund manager added genuine skill/value.

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Jensen's Alpha  =  Actual Return − [Rf + β × (Market Return − Rf)]
α_P  =  μ_P  −  [μ_f + β_P·(μ_mt − μ_f)]
```

Alpha Value	Interpretation	Action
$\alpha > 0$	Portfolio OUTPERFORMED CAPM — manager added genuine value	BUY / HOLD
$\alpha = 0$	Portfolio matched CAPM prediction — no excess value added	Neutral
$\alpha < 0$	Portfolio UNDERPERFORMED CAPM — manager destroyed value	SELL / SWITCH

Real-Life Example — Jensen's Alpha

Market return=14%, Risk-free=7%. Portfolio P: β=1.2, observed μ_P=18%.
CAPM expected = 7 + 1.2×(14−7) = 7 + 8.4 = 15.4%
Jensen's α = 18 − 15.4 = +2.6% → Manager added 2.6% alpha above CAPM expectation. ✓

Portfolio Q: β=0.8, observed μ_Q=12%.
CAPM expected = 7 + 0.8×7 = 12.6%
Jensen's α = 12 − 12.6 = −0.6% → Manager UNDERPERFORMED vs CAPM! ✗

This is why actively managed funds are compared to benchmark indices — alpha is what you pay management fees for!

Ratio	Risk Used	Best For	Formula

Sharpe Ratio	Total Risk (σ)	Evaluating stand-alone portfolios	S = (R−Rf) / σ
Treynor Ratio	Systematic Risk (β)	Comparing funds within a diversified portfolio	T = (R−Rf) / β
Jensen's Alpha	Beta-adjusted expected return	Measuring fund manager's skill	α = R − CAPM Expected R

UNIT III — Capital Market Theory & Pricing Models

17. Capital Market Line (CML)

Capital Market Theory extends the Markowitz framework by introducing a risk-free asset (like a Government T-Bill). When you add a risk-free asset, the new efficient frontier becomes a STRAIGHT LINE — the Capital Market Line.

The best line from the risk-free point, tangent to the Markowitz frontier, gives the CML:

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CML:   μ_P  =  μ_f  +  [(μ_mt − μ_f) / σ_mt]  ×  σ_P
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The tangency point is the Market Portfolio (σ_{mt}, μ_{mt}). The slope = Sharpe Ratio of the market = maximum return per unit of risk achievable.

What Does the CML Tell You?

Every efficient investor should hold a combination of:
1. The Market Portfolio (all risky assets in their market-cap proportions)
2. The Risk-Free Asset (like a T-Bill / Government Bond)

Conservative investors: more risk-free asset. Aggressive investors: borrow at risk-free rate to invest MORE in the market portfolio (leverage).

Real-Life Example — CML

Risk-free rate = 7%. Market portfolio: μ_{mt} = 14%, σ_{mt} = 18%.
CML: $\mu = 7 + [(14-7)/18] \times \sigma = 7 + 0.389 \cdot \sigma$

Conservative investor (σ=9%): $\mu = 7 + 0.389 \times 9 = 10.5\%$
→ Put 50% in T-bills, 50% in market portfolio.

Aggressive investor (σ=27%): $\mu = 7 + 0.389 \times 27 = 17.5\%$
→ Borrow at 7% and invest 150% in market portfolio (leveraged position).

18. One Fund Theorem (CAPM Context)

There exists a SINGLE portfolio of risky assets — the Market Portfolio M — such that every efficient portfolio is simply a combination of M and the risk-free asset.

In Simple Words

You don't need thousands of different portfolios. Just ONE: the Market Portfolio.

Mix it with the risk-free asset in different proportions to suit any investor's risk preference.

This is why passive index investing works so well!

The Market Portfolio

- Contains ALL risky assets, weighted by their market capitalisation.
- If any asset is excluded, it would have zero value — impossible in equilibrium.
- Each asset's weight = its market cap / total market cap.
- Beta of market portfolio = 1 by definition.

□ Real-Life Approximation

Nifty 50 or Sensex ≈ Market Portfolio for Indian investors.
A Nifty 50 index fund is 'close enough' to the theoretical market portfolio for most practical purposes.

19. Capital Asset Pricing Model (CAPM)

CAPM is the most important model in all of finance. It answers: 'For a given asset, what return should I EXPECT?' Expected return depends ONLY on the asset's sensitivity to the overall market — measured by Beta (β).

19.1 Beta (β) — The Measure of Systematic Risk

$$\beta\_i = \mathrm{Cov}(r\_i, r\_mt) / \sigma^2\_mt$$

Beta Value	Interpretation	Indian Example
β = 1	Asset moves exactly with the market	Nifty 50 Index Fund
β > 1	MORE volatile than market (aggressive)	Small-cap/startup stocks, Adani Green
β < 1	LESS volatile than market (defensive)	FMCG stocks, HUL, Nestle
β = 0	No correlation with market	Government T-bills, risk-free bonds
β < 0	Moves OPPOSITE to market — natural hedge!	Gold (in some crises), inverse ETFs

19.2 Security Market Line (SML) — The CAPM Equation

$$E(R_i) = Rf + \beta_i \times [E(R_{\square}) - Rf]$$

Where: Rf = Risk-free rate, β_i = Beta of asset i, $E(R_{\square})$ = Expected market return, $[E(R_{\square})-Rf]$ = Market Risk Premium

□ Key Concept — SML

- Any asset fairly priced by the market lies ON the SML.
- Asset plots ABOVE SML → undervalued → BUY signal.
- Asset plots BELOW SML → overvalued → SELL signal.
- Beta of a portfolio = weighted average of individual betas: $\beta\_P = \Sigma\, w_i \cdot \beta_i$

□ Real-Life Example — CAPM Pricing

Risk-free rate (Rf) = 6%. Market expected return = 14%. Market risk premium = 8%.

HDFC Bank with β = 0.8: Expected return = 6 + 0.8×8 = 12.4%
→ If HDFC Bank actually returns 14% → ABOVE SML → UNDERVALUED → BUY.

Adani Green with β = 1.5: Expected return = 6 + 1.5×8 = 18%
→ If Adani Green only returns 15% → BELOW SML → OVERVALUED → SELL.

CAPM tells you the FAIR return for the risk you're taking.

19.3 CAPM Assumptions

- All investors are rational and risk-averse.
- All investors have the same expectations about returns (homogeneous expectations).
- No taxes, transaction costs, or restrictions on short selling.
- Investors can borrow and lend at the risk-free rate.
- The market is perfectly competitive (no single investor affects prices).

CAPM Critique: These assumptions don't hold perfectly in real life. That's why extensions like APT and multi-factor models were developed. But CAPM remains the STARTING POINT of all asset pricing.

20. SML vs CML — Key Comparison

Feature	Security Market Line (SML)	Capital Market Line (CML)
X-axis	Beta (β) — systematic risk only	Standard Deviation (σ) — total risk
Applies To	ALL assets (individual stocks + portfolios)	Only EFFICIENT portfolios
Used For	Pricing: find if asset is over/underpriced	Portfolio construction
Slope	Market Risk Premium = E(R□)−Rf	Sharpe Ratio of Market
Equation	E(Ri) = Rf + β × (Rm − Rf)	μ_P = μ_f + [(μ_mt−μ_f)/σ_mt] × σ_P

Quick Memory Aid

SML → Pricing individual stocks: 'Is this stock cheap or expensive relative to its beta?'
CML → Constructing portfolios: 'What mix of risky + risk-free gives me the best trade-off?'

21. Arbitrage Pricing Theory (APT)

APT (Stephen Ross, 1976) is a more general alternative to CAPM. While CAPM uses ONE factor (market risk), APT allows MULTIPLE factors to drive returns.

APT Model

$$\text{Single factor:} \quad R_i = a_i + b_i \cdot f + \varepsilon_i$$

$$\text{Multi-factor:} \quad R_i = a_i + b_{1i}F_1 + b_{2i}F_2 + \dots + b_{\square i}F_{\square} + \varepsilon_i$$

Where: a_i = expected return component, b_i = factor sensitivity (like β in CAPM), f = factor (e.g., market return, GDP, inflation), ε_i = idiosyncratic (unsystematic) noise.

The No-Arbitrage Pricing Result

$$E(R_i) = \mu\_f + b_{i1} \cdot \lambda_1 + b_{i2} \cdot \lambda_2 + \dots + b_{i\square} \cdot \lambda_{\square}$$

Where $\lambda_{\square}$ = factor risk premium. CAPM is a special case of APT with one factor (market excess return) and λ = μ_mt − μ_f.

Feature	CAPM	APT
Factors	1 (Market only)	Multiple (Market, GDP, Inflation, Interest rates, etc.)
Foundation	Mean-Variance optimization	No-Arbitrage principle
Assumptions	Stricter (homogeneous expectations)	Looser, more realistic
Practical Use	Simple, widely used	More flexible, used in factor investing
Famous Example	Beta models (SEBI, Bloomberg)	Fama-French 3-Factor Model

Real-Life Example — Fama-French 3-Factor Model (APT Extension)

f_1 = Market premium (Rm − Rf) — like CAPM beta
f_2 = SMB = Small Minus Big (small-cap premium — small companies historically return more)
f_3 = HML = High Minus Low (value premium — high book-to-market firms return more)

$$r\_i = Rf + \beta_1 (Rm - Rf) + \beta_2 \cdot SMB + \beta_3 \cdot HML + \varepsilon_i$$

In India: Nifty Smallcap index historically outperformed Nifty 50 (positive SMB effect).
Value stocks like Coal India vs growth stocks like Zomato (HML effect).
This is why small-cap value funds often outperform — they load on factors 2 and 3!

22. Index Models — Single Factor & Multi-Factor

For large portfolios with many assets, computing ALL pairwise covariances is impractical. For 500 stocks, you need 500×499/2 = 124,750 covariances! Index models dramatically simplify this.

22.1 Single Index Model (Characteristic Line)

$$R_i - Rf = \alpha_i + \beta_i (R_{\square} - Rf) + \varepsilon_i$$

This is called the Characteristic Line (CL) of asset i. α_i is Jensen's alpha, β_i is systematic risk, ε_i is the unsystematic/idiosyncratic residual.

Key Assumptions of Single Index Model

- $E(\varepsilon_i) = 0$ for all assets (residuals have zero mean).
- $Cov(\varepsilon_i, R_{\square}) = 0$ (residual is uncorrelated with market return).
- $Cov(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$ (residuals of different assets are uncorrelated — KEY ASSUMPTION).

The last assumption is key: assets are only correlated THROUGH the market. This massively simplifies covariance computation:

$$\text{Covariance between assets: } \sigma_{ij} = \beta_i \times \beta_j \times \sigma^2\_m \quad (\text{for } i \neq j)$$

□ Real-Life Example — Why Stocks Move Together

Why do HDFC Bank and Reliance move together? Not because they're in the same business (they're not!) but because they both respond to the same market factor — Nifty's movements.

HDFC Bank: b_1 =0.9 (fairly tracks market). Reliance: b_2 =1.1 (slightly more volatile).
Cov(r_HDFC, r_Reliance) = 0.9 × 1.1 × σ²_Nifty

Instead of estimating millions of pairwise covariances, you only need n factor loadings and σ²_f.

22.2 Multi-Factor Model

When ONE factor (market) is insufficient, use multiple factors. Common factors include:

- Macro factors: GDP growth, inflation, interest rates, exchange rates
- Statistical factors: Principal components extracted from return data
- Firm characteristics: P/E ratio, dividend yield, book-to-market ratio

$$R_i = a_i + b_{i1} \cdot f_1 + b_{i2} \cdot f_2 + \ldots + b_{ik} \cdot f_k + \varepsilon_i$$

$$Cov(r_i, r_j) = b_{i1} \cdot b_{j1} \cdot \sigma^2_{f1} + (\text{cross-factor terms}) + b_{i2} \cdot b_{j2} \cdot \sigma^2_{f2}$$

CAPM as Special Case

CAPM is literally a single-factor model where the only factor is the market excess return.
APT and multi-factor models are GENERALISATIONS of CAPM.

As you add more factors, you explain more of the return variation — but also add complexity.

23. Absolute Deviation (AD) as a Risk Measure

Developed by Konno & Yamazaki (1991). Absolute Deviation (AD) is an alternative to variance that converts the quadratic optimization into a LINEAR program — much faster to solve.

$$AD(w) = E[\ |\Sigma_i\ w_i(r_i - \mu_i)|\ ] = (1/T)\ \Sigma_t\ |\Sigma_i\ w_i(r_{it} - \mu_i)|$$

Optimization Problem (Linear):

$$\text{Minimize} \quad (1/T)\ \Sigma_t\ y_t$$

$$\text{Subject to: } y_t \geq \Sigma_i\ w_i(r_{it} - \mu_i),\quad y_t \geq -\Sigma_i\ w_i(r_{it} - \mu_i),\quad \Sigma w_i = 1,\quad \Sigma w_i\mu_i \geq \mu_0$$

Real-Life Use of AD

Used by large fund houses managing thousands of assets where real-time rebalancing is needed.
The linear structure allows extremely fast computation using the Simplex method.
Drawback: AD doesn't capture the correlation structure as precisely as variance.

BONUS — Additional Important Topics

B1. Efficient Market Hypothesis (EMH)

EMH states that asset prices ALREADY reflect all available information. Therefore, you cannot consistently beat the market because any edge you think you have is already priced in.

Form	What is Priced In	Implication
Weak Form	Past price and volume data	Technical analysis cannot consistently work
Semi-Strong Form	All PUBLIC information (news, earnings, filings)	Fundamental analysis cannot consistently beat market
Strong Form	ALL information including insider information	Even insiders can't earn excess returns (generally too extreme)

Real-Life Example — EMH

If EMH is true, an index fund (which just holds the market) should beat most active fund managers. Empirically, over the long run, ~80-90% of active fund managers underperform their benchmark index. That's why Warren Buffett recommends index funds for ordinary investors!

B2. Risk-Return Tradeoff & Investor Utility

Different investors have different risk tolerances, captured by Utility Functions:

- Risk-Averse Investor: Requires extra return (risk premium) to accept more risk. Concave utility. MOST investors are risk-averse.
- Risk-Neutral Investor: Cares only about expected return, ignores risk. Linear utility.
- Risk-Seeking Investor: Actually enjoys risk. Convex utility. (Gamblers!)

Utility: $U = E(R) - (1/2)\cdot A\cdot\sigma^2$ where A = investor's risk aversion coefficient

Real-Life Example — Utility Function

Investor with A=4:
Portfolio X: Return=10%, σ=15%: $U = 10\% - 0.5\times4\times(15\%)^2 = 10\% - 4.5\% = 5.5\%$
Portfolio Y: Return=8%, σ=5%: $U = 8\% - 0.5\times4\times(5\%)^2 = 8\% - 0.5\% = 7.5\%$

The investor PREFERS Portfolio Y (lower return but MUCH safer).
This explains why many investors hold bonds even when stocks have higher expected returns!

B3. Beta Estimation & Regression

In practice, Beta is estimated by regressing an asset's historical returns against market returns. The slope of the regression line IS the beta.

$$R_i = \alpha + \beta \times R_{□} + \varepsilon \quad \text{(OLS Regression)}$$

□ Practical Tip

In Bloomberg, NSE/BSE terminals, or Yahoo Finance, you can find pre-calculated betas.
Remember: historical beta is NOT always a reliable predictor of future beta — it can change over time!
Typically, betas tend to revert towards 1 over long periods (Blume adjustment).

B4. Correlation vs Covariance

$$\text{Covariance:} \quad \sigma_{ij} = E[(r_i - \mu_i)(r_j - \mu_j)]$$

$$\text{Correlation:} \quad \rho_{ij} = \sigma_{ij} / (\sigma_i \times \sigma_j) \quad \text{[always between −1 and +1]}$$

□ Key Distinction

Covariance tells you the DIRECTION and MAGNITUDE of co-movement (units dependent on asset units).
Correlation is STANDARDISED — always between −1 and +1 — making it easier to interpret
and compare across different asset pairs.

B5. Portfolio Rebalancing

Over time, as asset prices change, portfolio weights drift away from their targets. Rebalancing means periodically buying/selling to restore target weights.

□ Real-Life Example — Rebalancing

You start with 60% Equity / 40% Bonds.
After a bull market, equities grow to 75% of portfolio.
Rebalancing means: SELL 15% of equities, BUY bonds to restore 60/40 split.

This automatically enforces 'buy low, sell high' discipline!

B6. Systematic Investment Plan (SIP) & Rupee Cost Averaging

Investing a fixed amount at regular intervals (SIP) automatically buys MORE units when prices are LOW and FEWER units when prices are HIGH. Over time this reduces the average cost per unit — called Rupee Cost Averaging.

Real-Life Example — SIP

You invest ₹5,000/month in Nifty Index Fund:
Month 1: NAV=₹100 → Buy 50 units
Month 2: NAV=₹80 → Buy 62.5 units ← More units for same money!
Month 3: NAV=₹110 → Buy 45.5 units

Average cost = ₹5000×3 / (50+62.5+45.5) = ₹95.1 per unit
Better than paying ₹100 upfront at the start!

B7. Leverage and Margin

Leverage means using borrowed money to invest. It amplifies both gains AND losses.

Real-Life Example — Leverage (Double-Edged Sword)

You have ₹1 lakh. You borrow ₹1 lakh at 8% and invest ₹2 lakh in Nifty.

If Nifty rises 15%: Portfolio = ₹2.3 lakh. Repay ₹1.08 lakh loan.
→ Left with ₹1.22 lakh. Return = +22% on your ₹1 lakh!

If Nifty falls 15%: Portfolio = ₹1.7 lakh. Repay ₹1.08 lakh loan.
→ Left with ₹0.62 lakh. Loss = −38% of your capital!

Leverage is a double-edged sword. Use with extreme caution.

Quick Reference Summary — All Key Formulas

Concept	Formula	Intuition
Return	$r = (V(T)-V(0)) / V(0)$	Profit % on your investment
Portfolio Return	$\mu_P = \Sigma w_i \mu_i = m^T w$	Weighted average of individual returns
Portfolio Variance	$\sigma^2_P = w^T C w$	Risk including all correlations between assets
Min Variance Portfolio	$w^* = C^{-1}e / (e^T C^{-1} e)$	Safest portfolio for given asset set
Efficient Frontier	Upper Markowitz Curve	Best risk-return combinations available
CML	$\mu = \mu_f + [(\mu_{mt}-\mu_f)/\sigma_{mt}]\cdot\sigma$	Blend risk-free + market portfolio
CAPM / SML	$\mu_i = \mu_f + \beta_i(\mu_{mt}-\mu_f)$	Fair return for systematic risk β
Beta	$\beta = Cov(r,r_{mt}) / \sigma^2_{mt}$	Sensitivity to market movements
Sharpe Ratio	$S = (\mu_P-\mu_f) / \sigma_P$	Return per unit total risk
Treynor Ratio	$T = (\mu_P-\mu_f) / \beta_P$	Return per unit systematic risk
Jensen's Alpha	$\alpha = \mu_P - [\mu_f+\beta(\mu_{mt}-\mu_f)]$	Value added by fund manager
APT	$E[r] = \mu_f + \Sigma b_{\square}\lambda_{\square}$	Multi-factor no-arbitrage pricing
MAD	$E[\lvert R - E(R)\rvert]$	Mean absolute deviation from expected return
VaR	Max loss at confidence α	Regulatory risk reporting standard
CVaR	Expected loss beyond VaR	Average loss in worst-case scenarios
MCR	$MCR_i = (Cw)_i / \sigma_P$	How much each asset contributes to portfolio risk

All the Best! All Syllabus Topics Covered ✓